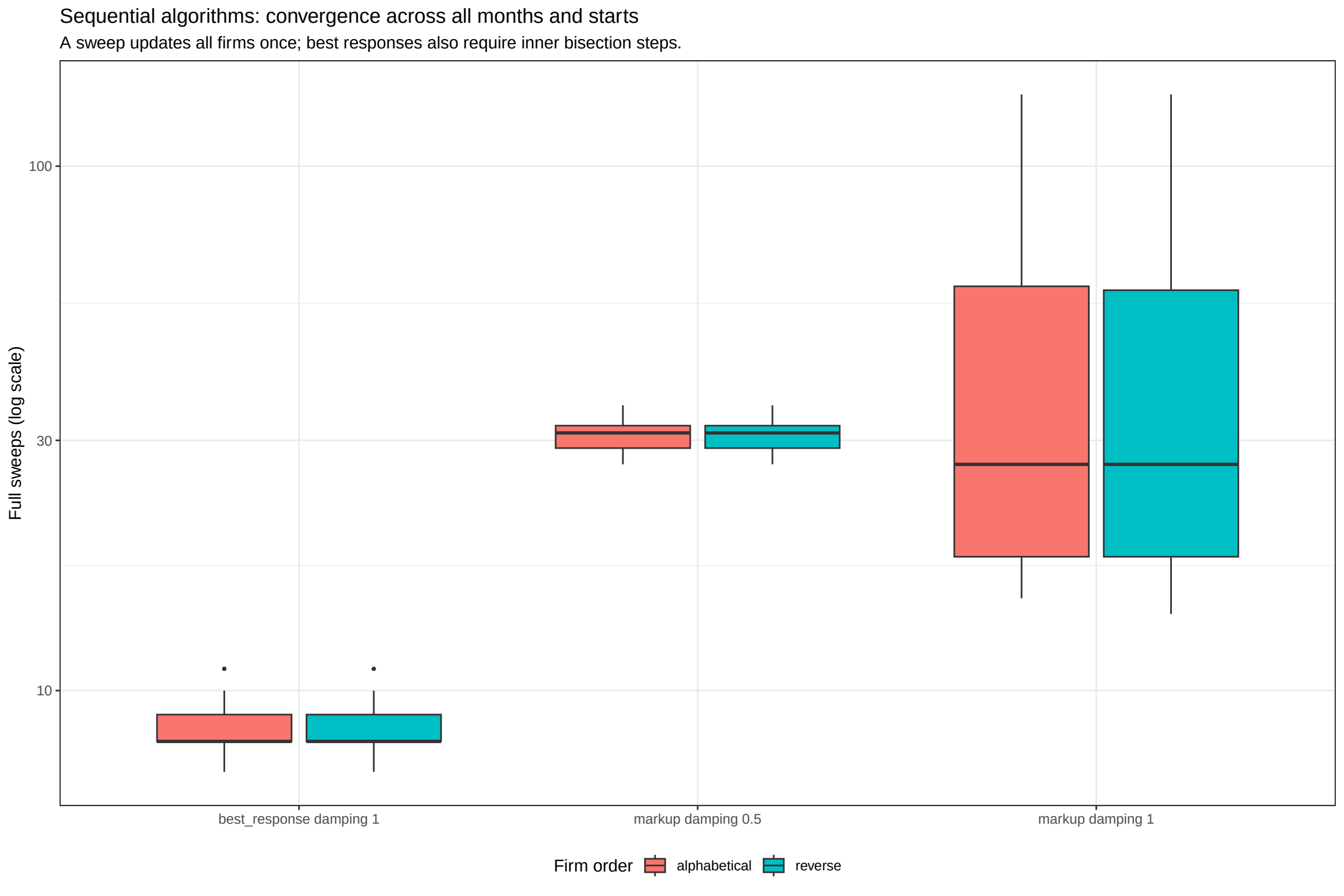
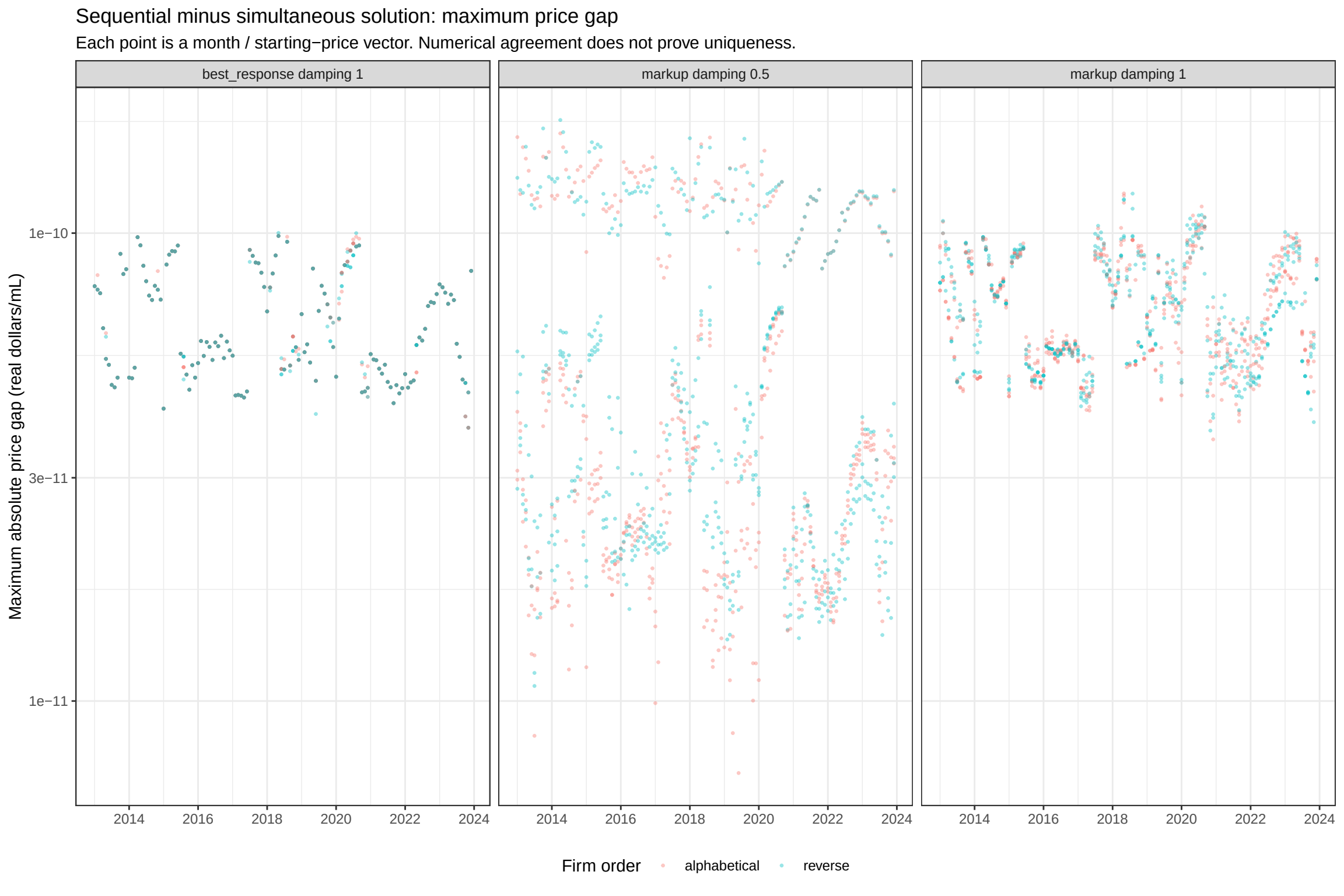
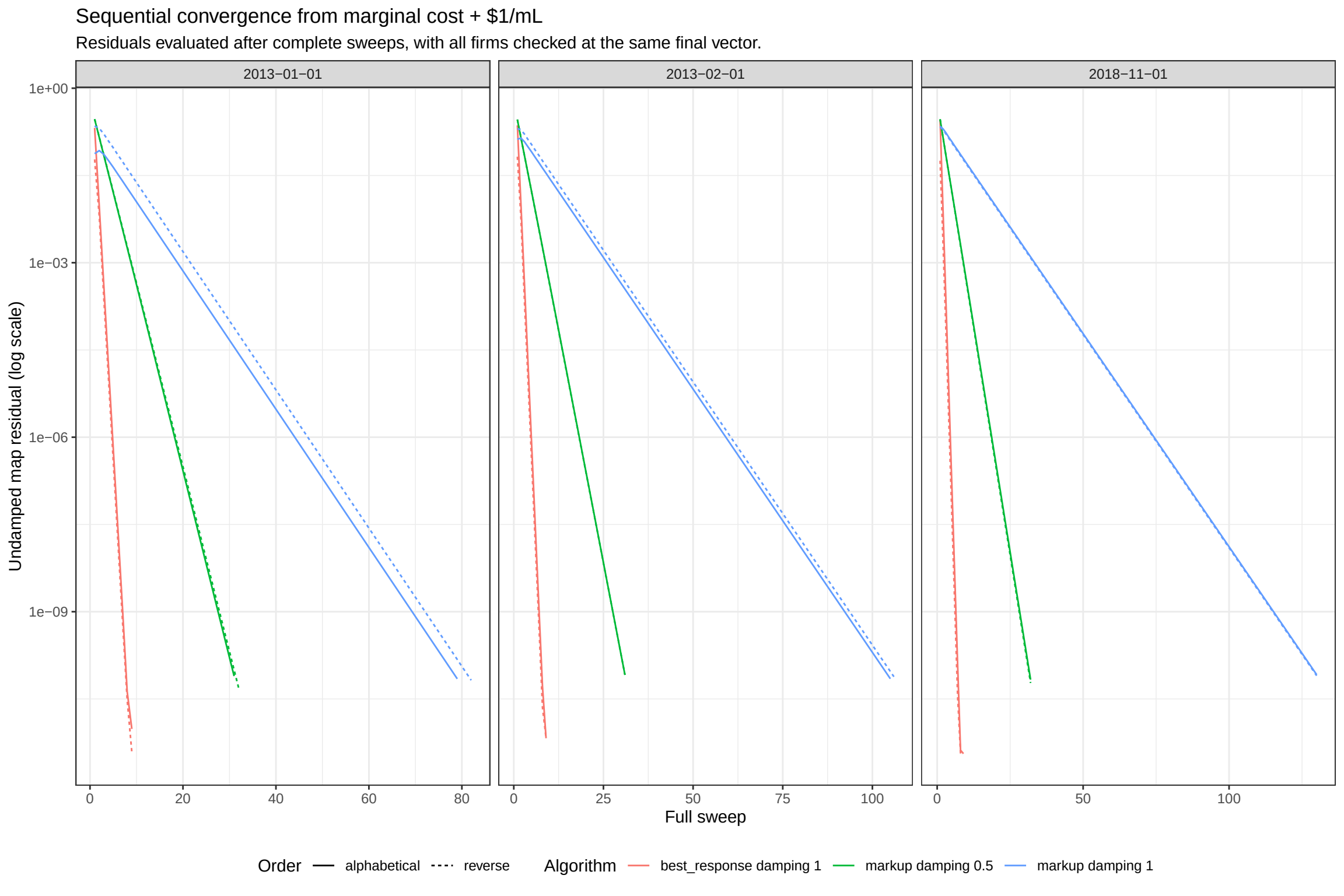

Sequential algorithms: convergence across all months and starts
A sweep updates all firms once; best responses also require inner bisection steps.
Full sweeps (log scale)
100
30
10
best_response damping 1
markup damping 0.5
markup damping 1
Firm order
alphabetical
reverse


Sequential minus simultaneous solution: maximum price gap
Each point is a month / starting−price vector. Numerical agreement does not prove uniqueness.
best_response damping 1
markup damping 0.5
markup damping 1
Maximum absolute price gap (real dollars/mL)
1e−10
3e−11
1e−11
2014
2016
2018
2020
2022
2024
Firm order
alphabetical
reverse


Sequential convergence from marginal cost + $1/mL
Residuals evaluated after complete sweeps, with all firms checked at the same final vector.
2013−01−01
2013−02−01
2018−11−01
Undamped map residual (log scale)
1e+00
1e−03
1e−06
1e−09
0
20
40
60
80
0
25
50
75
100
0
50
100
Full sweep
Order
alphabetical
reverse
Algorithm
best_response damping 1
markup damping 0.5
markup damping 1